Abhay Singh

Machine Learning Engineer

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PROFESSIONAL EXPERIENCE

Machine Learning Engineer, **O'Neill School Of Public And Environmental Affairs | Indiana, USA** **May 2025 - Present**

- Developed Bayesian and Gradient Boosted models to forecast chemical release volumes and site cleanup durations for 50k+ records. Built a geospatial clustering pipeline that improved PFAS hotspot identification by 9% over baseline.
- Architected a two-tier RAG system using FAISS and OpenAI embeddings to route queries between structured databases and LLM knowledge. Integrated NeMo Guardrails to ensure responses adhered to EPA validated safety standards.
- Orchestrated an Airflow ETL pipeline across 92 Indiana counties, engineering features from 10+ data sources to model toxicological exposure-response curves. Built a data framework to quantify and visualize premature mortality risks.

Machine Learning Engineer, **AP Moller Maersk | Bangalore, India** **Apr 2022 - Jun 2024**

- Achieved a 74% boost in routing efficiency and $200K annual savings by deploying an ensemble model microservice. This was accomplished by optimizing cost functions to provide real-time recommendations across global trade lanes.
- Reduced vessel travel time by 4% by engineering a container stacking optimization engine using K-Means and DBSCAN clustering integrated with Spark. Minimized "re-handles" by refining stacking logic across large-scale terminal datasets.
- Improved booking rates across 5 major trade lanes with 95% statistical confidence by executing rigorous A/B testing on a dynamic pricing feature. Designed experimental frameworks and analyzed results to ensure sample significance.
- Processed 500GB/day of IoT data via Kafka Streams and PySpark pipelines with automated quality checks, leveraging streaming frameworks and distributed computing concepts to drive a 45% reduction in shipment delays.

Data Engineer 2, **Microsoft | Hyderabad, India** **Apr 2019 - Apr 2022**

- Achieved 100% SLA compliance for terabyte-scale data by building concurrent Azure Data Factory and Databricks pipelines. Developed framework for 10+ business metrics to ensure high-impact data availability for executive reporting.
- Cut OLAP load times from 140s to 2s by optimizing DAX measures and building pre-aggregated fact tables using advanced SQL. Built result-caching via materialized views to offload heavy query workloads from Azure Synapse.
- Optimized refresh runtimes by 66% (3 hrs to 1 hr) and improved system throughput by offloading query workloads from Azure Synapse to distributed Scala-based pipelines with result caching via materialized views.
- Addressed 200+ critical support requests for Microsoft's Finance team, ensuring the accuracy and availability of high-impact business data used in executive reporting, audits, and operational planning.

EDUCATION

Indiana University, Bloomington, USA - *Master of Science, Data Science* **Aug 2024 - May 2026**

PROJECTS

PoseNet: Ape Pose Detection | Python, Computer Vision, Deep Learning, PyTorch, MMPose GitHub

- Attained 93.4% PCK@0.2 accuracy in primate pose estimation by fine-tuning ViTPose and ResNeXt models via supervised heat-map regression using the MMPose library on 70,000+ annotated images.

LLM Cognitive Fidelity Benchmark | Python, LLama 3.1, Langchain, Vector DB GitHub

- Benchmarked Llama 3.1 cognitive fidelity against 462 human subjects to evaluate reasoning alignment. Isolated false-positive intrusion patterns to optimize downstream fine-tuning and retention accuracy.
- Quantified model reliability through granular error analysis to identify cognitive biases and retrieval failures. Developed a framework to measure reasoning consistency compared to human performance baselines.

Customer Propensity & Retention Engine | Data Modelling, Feature Engineering, XGBoost, Light GBM GitHub

- Developed an E-commerce propensity engine achieving an 8% PR-AUC score despite a severe 3% class imbalance. Engineered an end-to-end pipeline with temporal splitting and instance-snapshots to prevent data leakage.
- Quantified purchase probabilities using Balanced Random Forest and XGBoost classifiers across massive transactional datasets. Leveraged PR-curves to capture minority "Repeat Buyer" segments, driving ad-targeting strategies.

SKILLS

Programming Languages : Python (NumPy, Pandas, Scikit-Learn, PyTest), C++, SQL, Java, Scala, PySpark, R

Machine Learning : Statistics, Regression, Classification, Decision Trees, KNN, XGBoost, LightGBM, Time-Series Forecasting

Gen AI : Large Language Models, RAG, Embeddings, Vector Databases, LangChain, GraphRAG

Cloud Computing : AWS, GCP, Azure, Snowflake, Docker, Kubernetes, GPU Optimization

Data Visualization : Power BI, Tableau, Looker, Matplotlib, Seaborn

Tools : Git, Postman, Linux, Unix, CUDA, Apache Airflow, Kafka, ETL/ELT tools

Deep Learning : PyTorch, Tensorflow, Keras, CNNs, Transformers, CUDA Programming

Analysis and Testing : A/B Testing, Statistical Significance, Hypothesis Testing, PR-AUC/F1 Optimization